

The Uneven Macroeconomic Effects of Carbon Markets¹

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Abstract

I present a macroeconomic assessment of the distributional impacts of carbon markets. Using a DSGE model with household heterogeneity calibrated to European Union data, I show that while carbon markets are effective in reducing emissions, an increase in the carbon price reduces production and wages and raises energy prices. Low-income households bear a disproportionate share of this burden, and both higher energy expenditures and lower labour income effects contribute to the gap in consumption response between household groups. Redistributing carbon revenues effectively offsets these uneven impacts without weakening emission reductions. Finally, I estimate that the share of revenues from the European carbon market allocated to supporting vulnerable households falls short of what would be needed to equalize the economic impact across households.

¹ Access to the data, Dynare and Python replication codes used for modelling and graphics are available on this [GitHub repository](#).

1 Introduction

Carbon markets, along carbon taxes, are considered by economists as the most efficient tools to address climate change. Yet they face strong public resistance when they raise energy and fuel prices as witnessed during the 2018 protests in France. Initiated in 2005, the European carbon market first targeted emissions from industry and power generation. In order to meet the goal of reducing emissions by 55 percent by 2030, the European Union aims to increase the pace of decarbonation by introducing in 2027 a new carbon market covering emissions from fuels used in road transport and heating. Extending the scope of the market to carbon-intensive commodities inevitably raises the question of how it will affect consumers.

The negative impact on consumer purchasing power and its uneven distribution across households is certainly what contribute the most to public discontent. The differences in the effects of carbon pricing across society, or distributional impacts, depend on several factors. First, households allocate different shares of their budgets to carbon-intensive goods, which are likely to experience the largest price increases. Second, households differ in their sources of income: some rely mainly on wages, while others can draw on savings or alternative financial income streams. Finally, regional differences in the energy mix affect the carbon intensity of certain commodities, particularly electricity, and therefore the pass-through of carbon pricing. Understanding how a carbon price increase propagates through the economy and who will bear the economic costs is crucial for macroeconomic policy and designing effective support for vulnerable households.

The objective of this work is to quantitatively assess the macroeconomic and distributional effects of a rise in carbon prices. I develop a Dynamic Stochastic General Equilibrium (DSGE) model with flexible prices and a climate block. Production sectors are covered by a carbon market and firms are required to purchase emission allowances. The model includes two different groups of households, heterogeneous in terms of their energy expenditure shares, income structure, and marginal propensities to consume (MPC). Under the carbon regulation, energy and non-energy producers abate their emissions until their marginal abatement costs equal the carbon price. The resulting costs of abatement and emissions allowances are passed on to energy prices and factor prices, such as wages and capital cost. Household groups with higher energy expenditure shares, or those heavily reliant on labour income due to limited savings or alternative revenue sources, are disproportionately affected. While higher energy bills account for most of the impact across all household groups, the indirect income channel disproportionately affects the level of consumption of financially constrained households.

Using the Household Budget Survey from Eurostat, I find substantial differences in the share of energy expenditures between low- and high-income households. By combining the survey with data on emissions by country and on commodities covered by the European carbon market, I identify the countries most vulnerable to a rise in carbon prices, which are those with both a high carbon intensity in these commodities and a high share of energy in household budgets. Concerning transport fuels, I find that the energy expenditures share are larger for the middle of the income distribution. It suggests that vulnerability to higher transport fuel prices depends less on income and more on road transport needs, which can vary depending on geographical factors such as living in an urban or rural area. Consider both vertical (across income groups) and horizontal (between regions and household characteristics other than income) dimensions is crucial when assessing the distributional effects of carbon pricing. I find that both aspects con-

tribute to differences in impact, but that heterogeneity in terms of income incidence of carbon pricing explains most of the gaps in consumption loss between households.

Redistributing part of the carbon revenues through targeted transfers to the most affected households can offset the regressive effects of carbon pricing and even make the system progressive without weakening emission reductions, since household energy demand is mostly inelastic. By calibrating the model to European Union data, I find that the share of revenues currently allocated to supporting vulnerable households is insufficient to fully equalise the economic impact across households groups. Given that people are generally averse to carbon taxes because they tend to overestimate their monetary losses (Douenne and Fabre [2022]), ensuring sufficient compensation and communicating on the equity of the transition is crucial for the public acceptance of the climate policy.

I first present the literature. In Section 2, I give context and data on distributive issues represented by carbon markets. In Section 3, I present a climate-economy general equilibrium model as well as its calibration to European Union data. In Section 4, I discuss the transmission channels of a carbon price shock and its distributional impacts, and evaluates redistribution policy scenarios. Finally, in Section 5, I discuss the policy implications of the results before concluding in Section 6.

Related literature

The traditional approach to studying the macroeconomic effects of climate policies relies on integrated assessment models (IAMs), which embed climate dynamics into a standard neoclassical growth framework (Nordhaus [1992]). These models evaluate the trade-off between the long-term economic costs and benefits of environmental policies. This approach has been widely used to compute the optimal carbon tax that internalizes the social cost of carbon (SCC), i.e. the economic damage caused by an additional ton of greenhouse gases (Nordhaus [2008], Golosov et al. [2014], Barrage [2019]). Estimating these damages is crucial for determining the optimal tax rate and remains an active area of research, with recent work suggesting higher SCC values (Bilal and Känzig [2024]). IAMs are also used to study the interaction between environmental policy and business cycles. Heutel [2012] finds that the optimal environmental policy should be procyclical. Sahuc et al. [2024] provides insights into the design of optimal monetary policy and inflation management during the transition to a low-carbon economy.

As practical experience with cap-and-trade systems and carbon taxes has grown, empirical research on carbon pricing effects has become increasingly rich and comprehensive. A large part of this literature focuses on the firm level, showing that carbon pricing is effective at reducing emissions but may not have a significant effect on firms' economic performance (Dechezleprêtre et al. [2019], Cui et al. [2019]). Recent work investigates the aggregate effects of carbon pricing, particularly within the EU ETS. Känzig [2023] exploits exogenous regulatory changes in the EU ETS to estimate the aggregate causal effects of carbon price shocks. The results show that restrictive carbon policies significantly reduce emissions but also entail short-run economic costs, including lower industrial production, higher unemployment, and increased energy and consumer prices. These effects are unevenly distributed, disproportionately affecting low-income households through both higher energy expenditures and greater income losses.

Beyond focusing solely on the importance of the economic costs, a large empirical literature studies the distributional effects of carbon pricing, i.e. how these effects are distributed across different groups in society. West and Williams [2004] documents the

regressivity of gasoline taxes, and emphasizes the crucial role of revenue recycling in mitigating this regressivity. In particular, flat recycling (i.e., redistributing revenues from the tax equally per person, regardless of income) could make a carbon tax progressive. [Bento et al. \[2009\]](#) analyses the effectiveness and distributional effects of US Gasoline taxes and finds that flat recycling largely offsets the regressivity of the tax, which increases the share of relatively fuel-efficient cars. [Williams et al. \[2014\]](#) shows that the incidence of a carbon tax depends primarily on how revenues are used, comparing alternatives such as capital tax cuts, lump-sum rebates, or labour tax reductions. [Pizer and Sexton \[2019\]](#) highlights the vertical regressive distributional effects of carbon pricing across the income distribution, but also stresses the importance of horizontal distributional effects, which occur among households with similar incomes but different characteristics, such as regional location, transportation access, or income sources. In the French carbon tax context, [Douenne \[2020\]](#) shows that commuting patterns, dwelling types, and energy use for heating and transport play an important role in shaping households' economic losses. More recent work on European Union data by [Bigio et al. \[2025\]](#) emphasizes indirect effects on labour income, finding that younger, poorer, and less-educated households, especially in Southern and Eastern Europe, bear a disproportionately high share of the economic costs of carbon pricing.

Alongside empirical research, a large body of work has been done in macroeconomics to capture the general equilibrium effects of climate policy. Using computable general equilibrium models, [Rausch et al. \[2014\]](#) highlights that carbon pricing affects households not only by raising the prices of carbon-intensive goods but also by changing factor prices (wages and interest rates) and [Goulder et al. \[2019\]](#) assess the direct and indirect effects of carbon taxes across income groups. Heterogeneous-Agent New Keynesian (HANK) models have also been employed to incorporate household income heterogeneity into climate policy analysis, as in [Bosetti and Maffezzoli \[2013\]](#). [Benmir and Roman \[2022\]](#) studies the transition dynamics toward a net-zero emissions target with a HANK model featuring a joint distribution of income and wealth. This thesis contributes to this literature by combining a Real Business Cycle (RBC) model with climate dynamics and household heterogeneity. Recent literature emphasized the role of MPC and income heterogeneity for the transmission of shocks ([Auclert \[2019\]](#)). The model uses the Two-Agent framework — simpler and more tractable than a fully heterogeneous-agent setup — originally developed by [Gali et al. \[2007\]](#) to account for the existing evidence of the effects of government spending on consumption, and then by [Bilbiie \[2008\]](#) to account for limited asset market participation in monetary policy transmission. [Chan et al. \[2024\]](#) uses a Two Agent New Keynesian model to study the transmission of energy price shocks.

2 Context and data

This section provides a brief overview of the functioning of global carbon markets and presents data on carbon prices. It also includes an analysis of the Eurostat Household Budget Survey (HBS) to examine how the share of household spending on fuels for transport and heating varies across income quintiles, and between countries with different carbon intensities of their energy mix.

About carbon markets

Carbon markets, also known as Emissions Trading Systems (ETS), are a market-based approach to controlling carbon emissions by providing economic incentives for

reducing greenhouse gas (GHG) emissions. They operate under a cap-and-trade principle: a cap is imposed on all emissions from covered installations, which declines over time to ensure continued emissions reductions. Companies receive emission allowances granting the right to emit one ton of CO₂ equivalent through auctions or free allocation. These allowances can then be traded on the market, thereby establishing a carbon price. Under this system, companies must monitor and report their emissions annually and surrender enough allowances to cover their emissions. Failure to comply with these requirements may result in heavy fines. Carbon allowances are storable and their purchase is not restricted to covered entities. Other market participants can buy allowances to resell them for profit in the future.

Following the commitments under the Kyoto Protocol, several groups of countries established emission trading systems. The four largest carbon markets today are the cap-and-trade programs of China, the European Union, Korea, and California. Figure 1 presents the evolution of carbon prices on these markets and shows that they have reached historic levels recently.

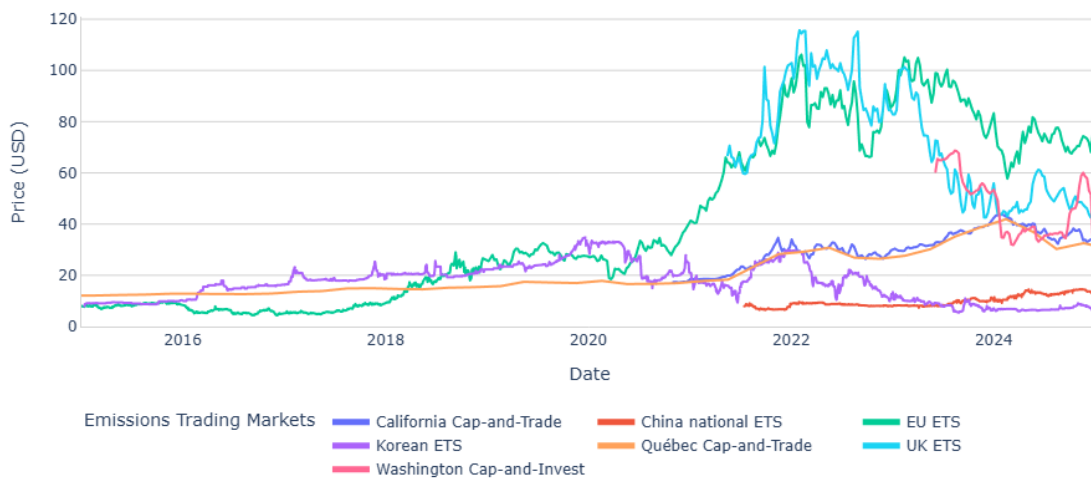


Figure 1: Carbon price trends in the main emissions trading systems

Notes: Carbon prices are the auction settlement prices of emission allowances for one ton of CO₂ equivalent. Allowance prices are not directly comparable across systems, given differences in ETS design. Source: ICAP

The EU ETS, established in 2005, currently has the highest carbon price among all systems. It covers CO₂ emissions from electricity and heat generation, energy-intensive industries, maritime transport, and aviation, together accounting for roughly 40% of the EU's GHG emissions. The current carbon price is approximately €76 per ton of CO₂. Starting in 2027, a second system, the EU ETS 2, will be introduced to cover emissions from fuels used in buildings, road transport, and small industries not already included in the existing ETS, together also representing about 40% of EU emissions. Under ETS 2, fuel suppliers rather than end consumers will be responsible for monitoring and reporting emissions.

Uncertainty remains regarding potential price trajectories and their underlying

drivers in both trading systems. Although the European Commission wants to set a cap of €45/tCO₂ for the ETS 2 carbon price, modeled price scenarios currently lie in the range of €71–261/tCO₂ (Günther et al. [2025]). Concerning the recycling of carbon market revenues, those are largely used to finance the energy transition. A minority portion of revenues is often allocated to redistribution measures, such as the European Union’s Social Climate Fund, to compensate the most affected households.

Before conducting a deeper analysis, I present an overview of the Household Budget Survey (HBS) from Eurostat to highlight differences in the consumption bundle of low- and high-income households, particularly with respect to energy expenditures for heating and transportation.

The Household Budget Survey

Before analyzing the distributional effects of carbon markets, it is necessary to review how the consumption bundles of low- and high-income households differ, particularly with respect to expenditures on energy for heating and transportation. I analyse HBS data on the structure of consumption expenditures by income quintile and consumption category purpose. Figure 2 shows that in 2022 the share of expenditures on electricity, gas, and other household fuels for heating was 8.48% for the first quintile and 4.83% for the fifth quintile. Transport-related expenses constitute a higher share of total expenditures for the highest income quintile, although this category combines fuel costs, vehicle purchases, and transport services. Variations in these shares for the first quintile may partly reflect the inclusion of additional countries in successive survey waves.

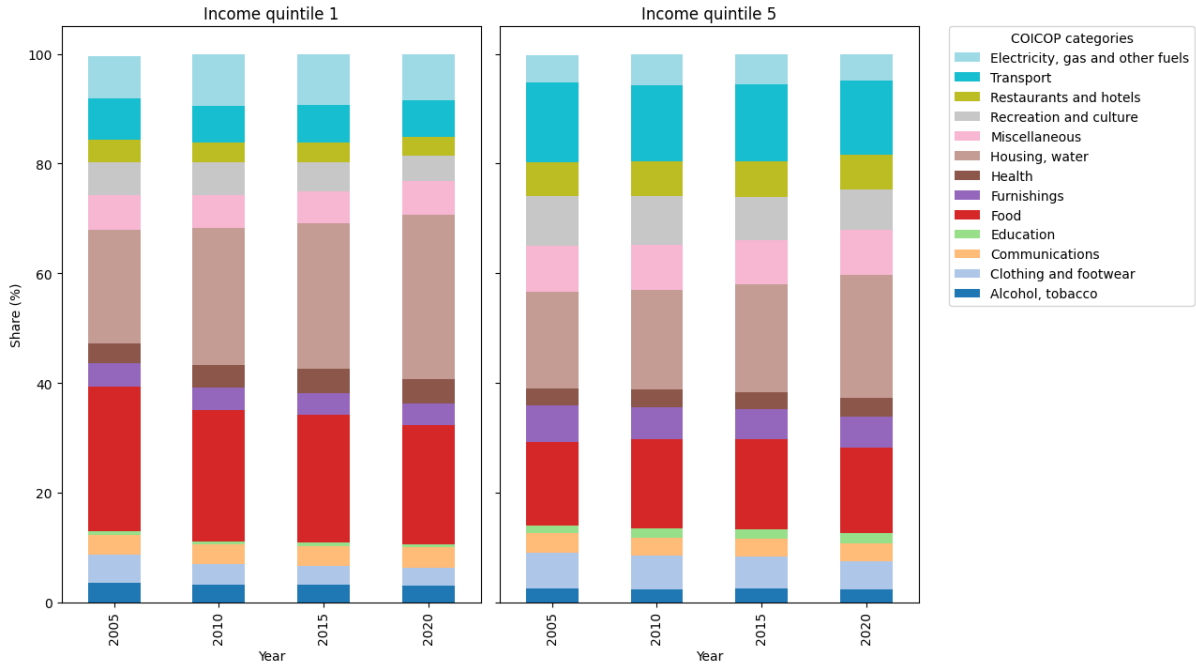


Figure 2: Consumption shares for low and high income households

Notes: Data for the European Union. Source: Eurostat

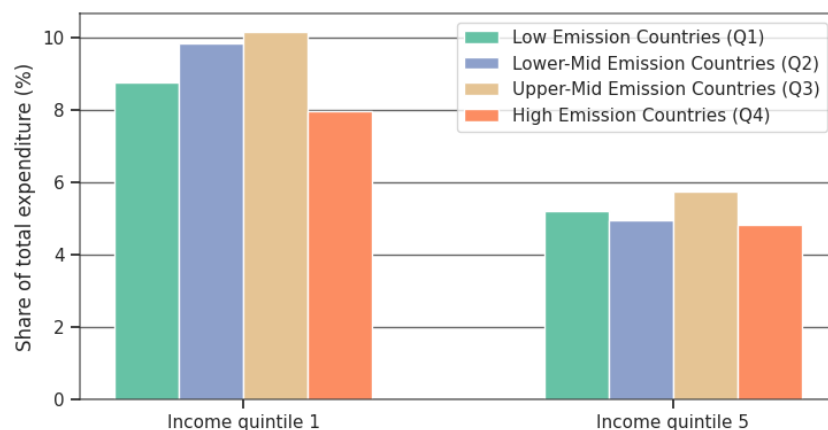


Figure 3: Household expenditure on heating energy (electricity, gas, liquid/solid heating fuels) by income quintile, grouped by emission level per household from ETS2-covered heating commodities

Notes: Data for the European Union in 2020 and Norway (data missing for Cyprus, France, Italy, Malta, Portugal, Czech Republic, Romania, Sweden). Source: Eurostat

Figure 3 further focuses on energy spending for heating for the first and the fifth income quintile, combining HBS data on household expenditures with emissions data from ETS-2-covered heating fuels. This makes it possible to classify countries according to their reliance on fossil fuel for heating purposes. Countries in the fourth emission per household quartile—including Poland, Germany, Ireland, Czechia, and Belgium—are those where heating fuel prices are most likely to rise if producers pass their cost increases on to consumers. Meanwhile, countries in the third quartile—including Austria, Croatia, Greece and Slovakia—exhibit the highest average share of household expenditure on heating energy, which makes their households particularly vulnerable to ETS 2 price increases.

Finally, figure 4 presents household spending on fuels for personal transportation. Interestingly, this share does not follow a monotonic pattern across income quintiles: although the first quintile spends relatively more than the fifth quintile on transport fuel, the highest shares are observed in the second and third quintiles. This finding suggests that vulnerability to higher transport fuel prices may depend less on income and more on transport needs and geographic factors. Rural households, which are more reliant on personal vehicles and have limited access to public transportation, may therefore be exposed to a greater financial burden from carbon pricing on transport fuels.

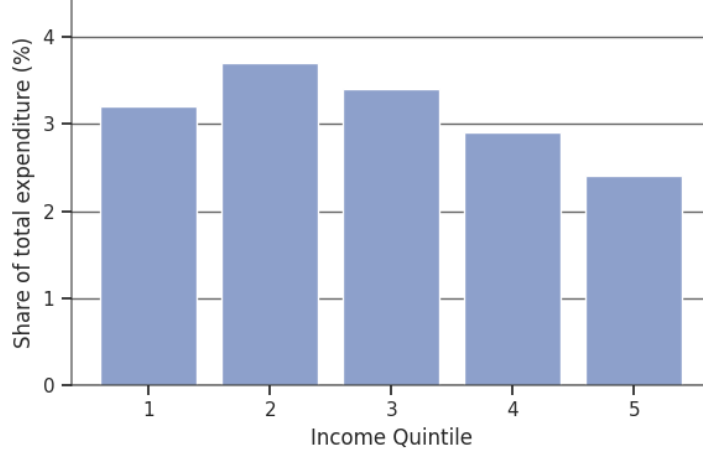


Figure 4: Household expenditure on fuels for personal transportation equipment, by income quintile

Notes: Data for Norway in 2010. Source: Eurostat

3 A climate-economy model

I build a model to formalize the idea that uneven consequences of carbon markets are driven by household heterogeneity in marginal propensities to consume (MPC), energy share in their consumption baskets, and their income composition. The model featuring a climate block belongs to the Dynamic Stochastic General Equilibrium (DSGE) class, and is similar in structure to [Känzig \[2023\]](#). The economy features two production sectors for energy and a non-energy consumption goods, two household groups and a government. The model is set in discrete time with an infinite horizon. Firm production in both sectors generates a negative externality through carbon emissions, that reduces the productivity of firms. To reduce emissions, the government requires firms to purchase carbon allowances for emissions they do not abate. Government finance public spending or fiscal transfers through the carbon market revenues.

3.1 Households

Using the same setup as [Gali et al. \[2007\]](#), I assume that there is a continuum of households $[0, 1]$ of two types: savers and hand-to-mouth. Households derive utility from consumption and experience disutility from labour supply.

3.1.1 Savers

Savers, indexed by S , constitute a share $1 - \omega$ of households. They are forward-looking Ricardian households, and smooth consumption with their savings. Each period t , a saver chooses an amount of consumption $C_{S,t}$, labour supply $L_{S,t}$ of labour supply, and next-period assets A_{t+1} solving the following intertemporal problem:

$$\max_{\{C_{S,t}, L_{S,t}\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t u(C_{S,t}, L_{S,t})$$

At time t , savers own a quantity A_t of risk-free assets which are claims on physical capital rented to firms at rate R_t . They earn labour income $W_t L_{S,t}$, receive dividends D_t from firms and transfers $T_{S,t}$ from the government. They purchase a consumption bundle at price $P_{S,t}$ a consumption bundle (its composition is detailed in Section 3.1.3) and invest I_t to accumulate assets. Their budget constraint at time t and law of motion for assets are:

$$\begin{aligned} P_{S,t} C_{S,t} + I_t &= R_t A_t + W_t L_{S,t} + D_t + T_{S,t} \\ A_{t+1} &= (1 - \delta) A_t + I_t \end{aligned}$$

Combining the two equations above gives the financial wealth of savers each period:

$$A_{t+1} = (1 + r_t) A_t + W_t L_{S,t} + D_t + T_{S,t} - P_{S,t} C_{S,t}$$

where $r_t = R_t - \delta$ denotes the risk-free rate (the user cost of capital net of depreciation) between t and $t+1$. The associated value function can be written as:

$$V_t(A_t) = \max_{A_{t+1}, L_{S,t}} u\left(\frac{1}{P_{S,t}}((1 + r_t)A_t + W_t L_{S,t} + D_t + T_{S,t} - A_{t+1}), L_{S,t}\right) + \beta V(A_{t+1})$$

First-order conditions and the envelope theorem give the Euler equation and the intratemporal labour-consumption condition. Together with the transversality (no Ponzi) condition, we obtain:

$$\begin{aligned} u_C(C_{S,t}, L_{S,t}) &= \beta(1 + r_{t+1}) \frac{P_{S,t}}{P_{S,t+1}} u_C(C_{S,t+1}, L_{S,t+1}) \\ -\frac{u_L(C_{S,t}, L_{S,t})}{u_C(C_{S,t}, L_{S,t})} &= \frac{W_t}{P_{S,t}} \\ \lim_{t \rightarrow \infty} \beta^t \frac{1}{P_{S,t}} u_C(C_{S,t}, L_{S,t}) A_{t+1} &= 0 \end{aligned}$$

3.1.2 Hand-to-Mouth households

Hand-to-mouth households, indexed by H , represent a share ω of households. They neither borrow nor save and are assumed to consume their entire current income each period. They earn wage income $W_t L_{H,t}$ and receive government transfers $T_{H,t}$, but do not receive dividends. Taking the price of their composite consumption bundle $P_{H,t}$ as given (see Section 3.1.3), they solve the following static problem:

$$\max_{C_{H,t}, L_{H,t}} u(C_{H,t}, L_{H,t}) \quad \text{s.t.} \quad P_{H,t} C_{H,t} = W_t L_{H,t} + T_{H,t}$$

The first-order conditions yield the intratemporal optimality condition which equates the marginal rate of substitution between leisure and consumption to the real wage:

$$-\frac{u_H(C_{H,t}, L_{H,t})}{u_C(C_{H,t}, L_{H,t})} = \frac{W_t}{P_{H,t}}$$

3.1.3 Consumption of goods and energy

The consumption bundle of households is composed of energy, denoted by E , and consumption goods, denoted by X . Following [Känzig \[2023\]](#), the consumption bundle of household $i \in \{H, S\}$ is a Constant Elasticity of Substitution (CES) aggregate of consumption goods and energy:

$$C_{i,t} = \left(a_{i,x}^{\frac{1}{\epsilon_x}} X_{i,t}^{\frac{\epsilon_x-1}{\epsilon_x}} + a_{i,e}^{\frac{1}{\epsilon_x}} E_{i,t}^{\frac{\epsilon_x-1}{\epsilon_x}} \right)^{\frac{\epsilon_x}{\epsilon_x-1}},$$

where ϵ_x is the elasticity of substitution between energy and consumption goods. Energy and consumption goods are perfect substitutes if ϵ_x approaches ∞ , and perfect complements if ϵ_x approaches 0.

Let $p_{x,t}$ and $p_{e,t}$ denote nominal prices of X and E , and $P_{x,t}$ and $P_{e,t}$ real prices. The nominal price index for the consumption bundle is $p_{i,t} = (a_{i,x} p_{x,t}^{1-\epsilon_x} + a_{i,e} p_{e,t}^{1-\epsilon_x})^{\frac{1}{1-\epsilon_x}}$. Choosing the non-energy good as the numéraire, the real price of the consumption good is $P_{x,t} = 1$ and the relative price of energy is $P_{e,t} = \frac{p_{e,t}}{p_{x,t}}$. The real price index is:

$$P_{i,t} = (a_{i,x} + a_{i,e} P_{e,t}^{1-\epsilon_x})^{\frac{1}{1-\epsilon_x}}$$

$a_{i,e}$ and $a_{i,x}$ are distribution parameters such that $a_{i,e} + a_{i,x} = 1$. They are calibrated to match the energy expenditure share of household i ,

$$s_i \equiv \frac{P_{e,t} E_{i,t}}{P_{i,t} C_{i,t}},$$

which implies

$$a_{i,e} = \frac{P_e E_i}{P_i C_i} \left(\frac{P_e}{P_i} \right)^{e_x-1} = s_i \left(\frac{P_e}{P_i} \right)^{e_x-1}$$

This specification ensures that the distribution parameters sum to one. Solving the intratemporal problem, detailed in [Appendix A.1](#), gives the demand schedule for energy and consumption goods:

$$\begin{aligned} X_{i,t} &= a_{i,x} \left(\frac{1}{P_{i,t}} \right)^{-\epsilon_x} C_{i,t} \\ E_{i,t} &= a_{i,e} \left(\frac{P_{e,t}}{P_{i,t}} \right)^{-\epsilon_x} C_{i,t}. \end{aligned}$$

3.2 Firms

The economy features two production sectors: an energy sector and a non-energy goods sector. Energy is used both directly in household consumption and as an input in intermediate goods production. Both sectors are covered by the carbon market and must pay a carbon price for their emissions. Labour and capital are perfectly mobile across sectors, so in equilibrium the wage and the rental rate of capital are equalized across sectors.

3.2.1 Energy producers

Energy producers supply energy E_t from labour force $L_{e,t}$ and capital $K_{e,t}$ using a constant return to scale (CES) technology. Output is given by the Cobb–Douglas production function:

$$E_t = A_{e,t}(K_{e,t})^{\alpha_e}(L_{e,t})^{1-\alpha_e}, \quad 0 < \alpha_e < 1.$$

where $A_{e,t}$ the Total Factor Productivity (TFP) in the energy sector and α_e the elasticity of energy production to capital.

Energy producers can provide a fraction $\mu_{e,t} \in [0, 1]$ of energy from zero-emissions technologies at an additional cost $F(\mu_{e,t}, E_t)$. Following the DICE specification of Nordhaus [2008], the abatement cost is increasing and convex in the amount of clean energy or abatement provided $\mu_{e,t}E_t$. Carbon emissions $\mathcal{E}_{e,t}(\mu_{e,t}, E_t)$ are the fraction of non-abated emissions from the energy production, which has an emission-intensity prior to abatement ϕ_e . Abatement cost and carbon emissions functions are:

$$\begin{aligned} F(\mu_{e,t}) &= f(\mu_{e,t})E_t = \theta_1(\mu_{e,t})^{\theta_2}E_t \\ \mathcal{E}_{e,t} &= (1 - \mu_{e,t})\phi_e E_t \end{aligned}$$

Real revenue for an energy producer is $P_{e,t}E_t$. Costs arise from wage payments $W_tL_{e,t}$, the rental cost of capital $R_tK_{e,t}$, abatement expenditures $F(\mu_{e,t}, E_t)$, and carbon allowance payments $\tau_t\mathcal{E}_{e,t}$ where τ_t is the carbon price. Profits from energy production are :

$$\Pi_{e,t} = P_{e,t}E_t - W_tL_{e,t} - R_tK_{e,t} - F(\mu_{e,t}, E_t) - \mathcal{E}_{e,t}(\mu_{e,t}, E_t)\tau_t$$

Taking $(P_{e,t}, W_t, R_t, \tau_t)$ as given, the energy producer chooses $K_{e,t}$, $L_{e,t}$, and $\mu_{e,t}$ to maximize profits. The first-order conditions are:

$$\begin{aligned} W_t &= \varrho_{e,t}(1 - \alpha_e)\frac{E_t}{L_{e,t}} \\ R_t &= \varrho_{e,t}\alpha_e\frac{E_t}{K_{e,t}} \end{aligned}$$

where $\varrho_{e,t}$ is the marginal cost of labour and capital inputs. Under perfect competition and constant returns to scale, the marginal revenue of energy production equate the marginal costs of abatement and carbon allowance and the marginal cost of labour and capital inputs, implying zero profits:

$$\varrho_{e,t} = P_{e,t} - f(\mu_{e,t}) - \phi_{e,t}(1 - \mu_{e,t})\tau_t$$

The optimal abatement condition is:

$$\tau_t = \frac{f'(\mu_{e,t})}{\phi_e}$$

specifying that abatement is chosen until the marginal abatement cost per unit of emissions avoided equals the carbon price.

In Appendix A.2, I develop the case where the level of abatement is partially rigid

and firms cannot fully adjust their abatement each period. Instead, an adjustment rule governs the effective aggregate abatement: $\mu_{e,t} = \lambda \mu_{e,t}^\# + (1 - \lambda) \mu_{e,t}$, where λ is the share of firms that reset their abatement and $\mu_{e,t}^\#$ their chosen abatement level. The effects of this rigidity are discussed in section 4.1.

3.2.2 Non-energy producers

The non-energy sector is splitted into two subsectors: a representative competitive final-goods firm that aggregates intermediate varieties using a CES technology, and a continuum of monopolistically competitive intermediate-goods producers that use capital, energy, and labour as inputs. Intermediate goods are imperfect substitutes and the demand for each variety decreases with its relative price. Prices are perfectly flexible and each intermediate firm sets its price as a constant markup over the marginal cost of production. Profits generated by intermediate good firms are rebated to savers as dividends.

Final good producers

The representative final-goods firm bundles a continuum of differentiated intermediate goods into an output of final goods using a CES technology. Taking the set of intermediate prices $\{p_t(i)\}_{i \in [0,1]}$ as given, it chooses quantities $\{y_t(i)\}_{i \in [0,1]}$ to maximize profits:

$$\begin{aligned} \max_{y_t(i)} \quad & P_t Y_t - \int_0^1 y_t(i) p_t(i) di \\ \text{s.t.} \quad & Y_t = \int_0^1 (y_t(i))^{\frac{\epsilon_y - 1}{\epsilon_y}} di)^{\frac{\epsilon_y}{\epsilon_y - 1}} \end{aligned}$$

where Y_t is the aggregate demand, and ϵ_y is the elasticity of substitution across varieties.

The first-order conditions give the demand function for each variety, which is downward-sloping in the relative price $p_t(i)/P_t$:

$$y_t(i) = \left(\frac{p_t(i)}{P_t} \right)^{-\epsilon_y} Y_t$$

Under perfect competition, the associated CES price index is

$$P_t = \left(\int_0^1 p_t(i)^{1 - \epsilon_y} di \right)^{\frac{1}{1 - \epsilon_y}}$$

Intermediate producers

Intermediate goods producers $i \in (0, 1)$ operate under monopolistic competition and supply differentiated varieties $y_t(i)$. Production uses labour $L_{y,t}(i)$, capital $K_{y,t}(i)$, and energy inputs $E_{y,t}(i)$, combined through a CES Cobb–Douglas technology:

$$y_t(i) = A_{y,t} d(s_t) K_{y,t}(i)^\alpha E_{y,t}(i)^\nu L_{y,t}(i)^{1 - \alpha - \nu}, \quad \alpha, \nu > 0, \quad \alpha + \nu < 1.$$

where α and ν measure the output elasticity of capital and energy, A_t is the TFP and $d(s_t)$ captures the effect of climate damages on productivity, modelled as an expo-

nential decay function of the atmospheric carbon stock $d(s_t) = e^{-\gamma s_t}$ as in Känzig [2023].

Similarly to energy producers, each producer can deliver a fraction $\mu_{y,t}(i)$ of output using clean (abated) technologies at an additional cost $F(\mu_{y,t}(i), y_t(i))$. Emissions $\mathcal{E}_{y,t}(i)$ are modelled as the share of unabated emissions from production, which has an emission intensity ϕ_y . The abatement cost and emission functions are:

$$\begin{aligned} F(\mu_{y,t}(i), y_t(i)) &= f(\mu_{y,t})y_t(i) = \theta_1(\mu_{y,t})^{\theta_2}y_t(i) \\ \mathcal{E}_{y,t}(i) &= (1 - \mu_{y,t}(i))\phi_y y_t(i) \end{aligned}$$

Firms maximise real profits, taking the production function and the demand schedule as given. Revenues are the real value of intermediate goods sales $\frac{p_t(i)}{P_t}y_t(i)$. Costs arise from renting capital $K_{y,t}(i)$ at rate R_t , employing labour $L_{y,t}(i)$ at wage W_t , purchasing energy $E_{y,t}(i)$ at price $P_{e,t}$, paying abatement costs, and purchasing emissions allowances at carbon price τ_t . Profits of intermediate goods producers are given by:

$$\begin{aligned} \Pi_{y,t}(i) &= \frac{p_t(i)}{P_t}y_t(i) - W_t L_{y,t}(i) - R_t K_{y,t}(i) - P_{e,t} E_{y,t}(i) - F(\mu_{y,t}) - \mathcal{E}_{y,t}\tau_t \\ &= \left(\frac{p_t(i)}{P_t} - mc_t(i) \right) y_t(i) \end{aligned}$$

where $mc_t(i)$ is the total marginal cost of production.

As monopolistic competitors, firms choose input demands and the abatement rate $\mu_{y,t}(i)$ to maximize profits, taking the downward-sloping demand schedule from the final-good producer as given. I solve the intermediate firms profit maximization problem in Appendix A.3. Since all intermediate goods producers face the same factor prices and share the same technology, optimality conditions are symmetric. We can write the FOCs for the intermediate-good sector as²:

$$\begin{aligned} R_t &= \varrho_{y,t} \alpha \frac{y_t}{K_{y,t}} \\ P_{e,t} &= \varrho_{y,t} \nu \frac{y_t}{E_{y,t}} \\ W_t &= \varrho_{y,t} (1 - \nu - \alpha) \frac{y_t}{L_{y,t}} \\ \tau_t &= \frac{f'(\mu_{y,t})}{\phi_y} \end{aligned}$$

where y_t is the aggregate production of intermediate goods and $\varrho_{y,t}$ is the marginal input cost, net of abatement and carbon price. The total marginal cost, identical across firms, is the sum of input cost, abatement and carbon allowance costs:

$$mc_t(i) = mc_t = \varrho_{y,t} + f(\mu_{y,t}) + (1 - \mu_{y,t})\phi_y \tau_t$$

I assume perfectly flexible prices. The only distortion in this framework arises from monopolistic competition and the ability for firms to set their price equal to a constant markup over their current marginal cost. As detailed in appendix A.3, the marginal cost

²As with the energy sector, I examine the case of frictions on the adjustment of the abatement level in section 4.1.

and dividends (equal to the firm's profits) are as follows:

$$mc_t = \frac{\epsilon_y - 1}{\epsilon_y}$$

$$d_t = (1 - mc_t)Y_t$$

3.3 Climate dynamics

The model features a reduced-form climate block following Golosov et al. [2014]. The aggregate flow of emissions is the sum of emissions of the energy and non-energy sectors: $\mathcal{E}_t = \mathcal{E}_{y,t} + \mathcal{E}_{e,t}$. The current level of atmospheric carbon concentration is a function of current and past emissions :

$$s_t = (1 - \phi)s_{t-1} + \phi_0\mathcal{E}_t$$

where ϕ_0 captures the share of emissions that do not immediately exit the atmosphere, and $1 - \phi$ measures how emissions decay over time.

3.4 Government

The government finances government spending and targeted transfers using carbon credit revenues. The share of carbon revenues attributed to transfers is determined by a parameter t_0 :

$$G_t + T_t = \tau_t\mathcal{E}_t$$

$$T_t = t_0\tau_t\mathcal{E}_t$$

The distribution of tax revenues are governed by the parameter μ , the share of transfers paid to hand-to-mouth households:

$$\omega T_{H,t} = \mu T_t$$

$$(1 - \omega)T_{S,t} = (1 - \mu)T_t$$

The carbon price is fixed at its steady-state level $\bar{\tau}$. In the simulations, the carbon price varies around $\bar{\tau}$ through a shock following an AR(1) process:

$$\tau_t = \bar{\tau}\epsilon_{\tau,t}, \quad \log(\epsilon_{\tau,t}) = \rho_{\tau}\log(\epsilon_{\tau,t-1}) + \eta_{\tau}$$

3.5 Aggregation and competitive equilibrium

Aggregate total, non-energy, and energy consumption are given by $C_t = \omega C_{H,t} + (1 - \omega)C_{S,t}$, $X_t = \omega X_{H,t} + (1 - \omega)X_{S,t}$ and $E_{C,t} = \omega E_{H,t} + (1 - \omega)E_{S,t}$. Because only savers hold assets, the aggregate capital level is $K_t = (1 - \omega)A_t$, and the aggregate investment is $I_t^{agg} = K_{t+1} - (1 - \delta)K_t$. The aggregate income is given by $Y^{agg} = X_t + P_{e,t}E_{C,t} + I_t^{agg} + G_t$.

A competitive equilibrium is defined as a sequence $\{C_{H,t}, C_{S,t}, X_{H,t}, X_{S,t}, E_{H,t}, E_{S,t}, L_{H,t}, L_{S,t}, Y_t, E_t, A_t, K_{t+1}, K_{y,t}, K_{e,t}, L_{y,t}, L_{e,t}, E_{y,t}, \mu_{y,t}, \mu_{e,t}, \mathcal{E}_{y,t}, \mathcal{E}_{e,t}, W_t, R_t, P_t, P_{S,t}, P_{H,t}, P_{e,t}, d_t, \tau_t\}_{t=0}^{\infty}$ such that:

- $\{C_{H,t}, C_{S,t}, X_{H,t}, X_{S,t}, E_{H,t}, E_{S,t}, A_t, L_{H,t}, L_{S,t}\}_{t=0}^{\infty}$ solves the households problem given and $\{W_t, R_t, P_{e,t}\}_{t=0}^{\infty}$

- $\{Y_t, K_{y,t}, K_{e,t}, L_{y,t}, L_{e,t}, E_{y,t}, \mu_{y,t}, \mu_{e,t}\}_{t=0}^{\infty}$ solves both producer's problem given $\{W_t, R_t, P_{e,t}, \tau_t\}_{t=0}^{\infty}$
- At $\{W_t, R_t, P_{e,t}, \tau_t\}_{t=0}^{\infty}$, all market clears:
 - $Y_t = X_t + I_t^{agg} + G_t + \sum_{s \in \{y,e\}} F(\mu_{s,t})$
 - $K_{y,t} + K_{e,t} = K_t$
 - $L_{y,t} + L_{e,t} = L_t$
 - $E_t = E_{y,t} + E_{C,t}$

3.6 Calibration

The model is calibrated using European Union data at a quarterly frequency. Household utility function $u(\cdot)$ has a Constant Relative Risk Aversion (CRRA) form:

$$u(C_{i,t}) = \frac{C_{i,t}^{1-\sigma}}{1-\sigma} - \chi_i \frac{L_{i,t}^{1+\theta}}{1+\theta}, \quad i \in \{S, H\}$$

where $\frac{1}{\sigma}$ is the intertemporal elasticity of substitution and $\frac{1}{\theta}$ is the Frisch labour supply elasticity.

Standard parameters values follow the macroeconomic literature. The discount factor β is set to 0.99, implying a risk-free rate of 1 percent. The capital depreciation rate δ is calibrated at 2.5 percent quarterly, yielding a user cost of capital at 3.5 percent. I set the intertemporal elasticity of substitution at two thirds and the Frisch elasticity at 0.78 as in [Barrage \[2019\]](#). The disutility-of-labour parameters χ_i are chosen so that steady-state share of hours worked L_i equal one third. The share of hand-to-mouth households ω is 20 percent, which corresponds to the first-quintile threshold in the Eurostat survey and is close to estimates of the share of poor hand-to-mouth households of [Kaplan et al. \[2014\]](#). The distribution parameters $a_{S,e}$ and $a_{H,e}$ are calibrated to match household expenditure shares on energy of 11.58 percent for hand-to-mouth households and 7.24 percent for savers³ in the Eurostat survey presented in Section 2. The elasticity of substitution between energy and non-energy goods is set to 0.2 following [Känzig \[2023\]](#).

On the production side, the capital output elasticity α and energy output elasticity ν in the non-energy sector are set to 0.3 and 0.03 respectively as in [Golosov et al. \[2014\]](#). TFP in the non-energy sector is set to 3.7 and to 1 in the energy sector. The capital output elasticity in the energy sector α_e is 0.6, according to the estimate in [Barrage \[2019\]](#). The elasticity of substitution across differentiated varieties is 6, which is, a standard value that implies a steady-state markup of 20 percent.

I use EU data provided by Eurostat to calibrate the emissions-intensity parameters and the carbon-allowance price. Table 1 reports the emissions and output shares for the energy and non-energy sectors. Given that 75 percent of emissions are covered by the European Emission Trading System (ETS), I set the aggregate emissions intensity, i.e the ratio of aggregate emissions to aggregate income, at 0.15. I then choose the sectoral intensities for the energy sector ϕ_e and the non-energy sector ϕ_y to be consistent with this aggregate intensity and the observed emissions share of both sectors⁴. I set

³Mean value observed in the European Union for heating energy and in Norway for transport fuel.

⁴Let s_e denote the energy-sector share of total emissions, θ_e its output share, and ϕ the aggregate emission intensity. The sectoral intensities satisfy $\phi = \theta_e \phi_e + (1 - \theta_e) \phi_y$ and $\phi_e = \frac{s_e \phi}{\theta_e}$

the steady-state carbon price $\bar{\tau}$ to 0.058. This value matches an implied carbon-revenue share of 0.87 percent of steady-state output, based on allowance prices of 75 euros in the ETS (covering 35% of emissions) and 45 euros in ETS 2 (covering 40%)⁵. Finally, the abatement-cost parameters θ_1 and θ_2 , for the two sectors are borrowed from Nordhaus [2008].

	Energy sector	Non-energy sector
Emissions share	44.7 %	55.3 %
Output share	1.77 %	98.23 %

Table 1: Emissions and output shares by sector

Notes: The energy-sector share of emissions is computed as the ratio of emissions from the energy sector (NACE Rev. 2 category D) plus direct household emissions to total EU emissions in the first quarter of 2025. The energy-sector share of output is the share of gross value added accounted for by NACE D category in 2023. Source: Eurostat

For the climate block, I follow Golosov et al. [2014], as adapted by Känzig [2023] to calibrate the climate damage parameter associated with atmospheric carbon concentration. Turning to the carbon cycle, I use the same values as Känzig [2023] and set the emission decay rate $1 - \phi$ at 0.9994 the fraction of emissions that remain in the atmosphere upon release ϕ_0 at 0.5359.

⁵Revenues are calculated based on CO2 emissions equivalent to 900 million tons, which is the level observed in the first quarter of 2025 in the European Union according to Eurostat

Table 2: Summary of key calibration parameters

Parameter	Description	Value	Source
β	Discount factor	0.99	Standard value
$1/\sigma$	Intertemporal elasticity of substitution	1	Standard value
$1/\theta$	Frisch elasticity of labour supply	1.28	Barrage [2019]
δ	Depreciation rate	0.025	Standard value
α	Capital output elasticity	0.3	Barrage [2019]
ν	Energy output elasticity	0.03	Barrage [2019]
α_e	Capital output elasticity (energy sector)	0.6	Barrage [2019]
ω	Share of hand-to-mouth	0.2	Eurostat threshold
$a_{H,e}$	Distribution parameter H	0.161	Energy share of 11.58%
$a_{S,e}$	Distribution parameter S	0.103	Energy share of 7.24%
ϵ_x	Elasticity of substitution between energy and goods	0.2	Känzig [2023]
ϵ_p	Price elasticity	6	Steady-state markup of 20%
γ	Climate damage parameter	5.3×10^{-5}	Golosov et al. [2014]
ϕ_0	Emissions staying in atmosphere	0.5359	Känzig [2023]
$1 - \phi$	Emissions decay parameter	0.9994	Känzig [2023]
$\bar{\tau}$	Steady-state carbon price	0.0587	Implied price from EU ETS
ϕ_e	Emissions intensity (energy sector)	0.98	Eurostat
ϕ_y	Emissions intensity (non-energy sector)	0.084	Eurostat
θ_1	Abatement cost parameter	0.1	Nordhaus [2008]
θ_2	Abatement cost parameter	2.7	Nordhaus [2008]
ρ_τ	Persistence of carbon tax shock	0.6	Känzig [2023]

4 Quantitative analysis

A key finding of this quantitative analysis is that a carbon price increase affects the consumption of low-income households more than twice as much as that of higher-income households. Household heterogeneity in the income incidence of carbon pricing explains two-thirds of the gap in consumption losses, while heterogeneity in energy-expenditure shares explains the remaining one-third. In addition, I estimate the share of distributional impacts arising from direct effects on energy prices and indirect effects on income. Both of these effects could be offset by targeted transfers. Redistribution schemes avoid the regressivity of carbon pricing without significantly compromising emissions reductions in the market.

Section 4.1 presents the macroeconomic dynamics triggered by a carbon price shock. Section 4.2 discusses the distributional effects of an increase in the carbon price. Section 4.3 highlights the role of heterogeneity in terms of energy expenditure and income incidence in shaping response gaps between households, and Section 4.4 shows that redistributing part of the revenues generated by the market through targeted transfers can offset the regressivity of the carbon price.

4.1 Carbon price shock and macroeconomic dynamics

How does an increase in the carbon price transmit through the economy? In this section, I examine how a 1% shock in the carbon price affects the main macroeconomic aggregates. Figure 5 displays the shock impulse responses.

When the carbon price increases by 1%, firms are forced to engage in more abatement efforts until the marginal abatement cost equals the carbon price. An increase in abatement and in the amount of allowances purchased raises the marginal cost of production for both firms. For energy firms, this ultimately raises the energy price by 0.45%. As a result, both households and firms decrease their energy demand, which forces a reduction in energy production by 0.24%. Firms also reduce demand for capital and labour inputs. Consequently, wages decrease by 0.05%, and the interest rate by 0.1%. The reduction in the interest rate reduces investment incentives for households. Asset holdings and the aggregate level of capital decline until the interest rate again reaches positive values. The overall effect on final-goods output is a reduction of 0.01%. It is driven by lower aggregate demand and a decline in aggregate labour supply caused by lower wages.

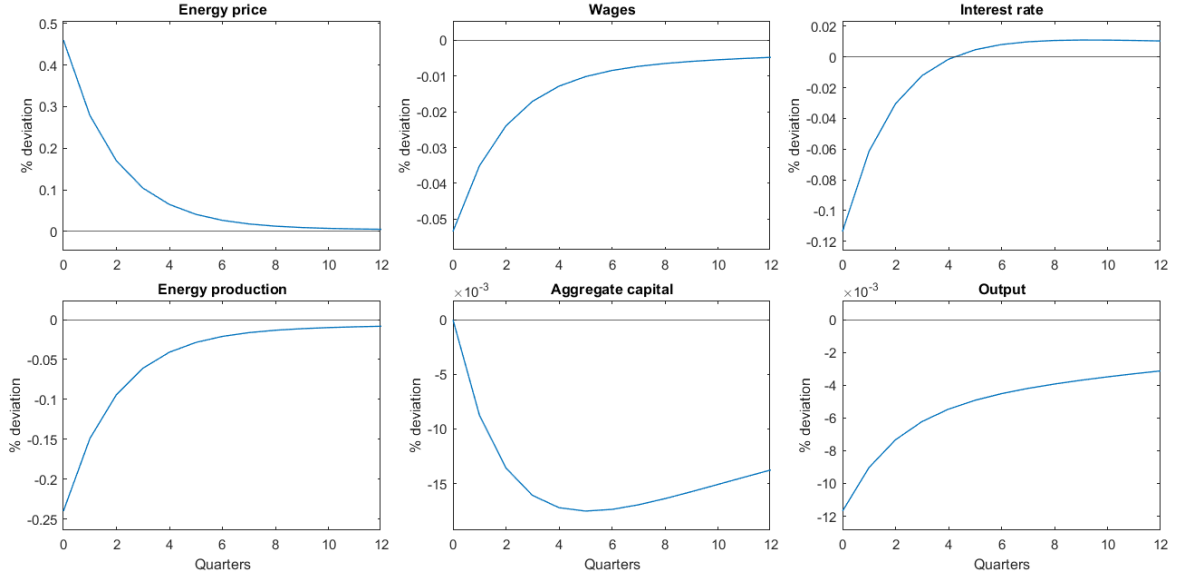


Figure 5: Firms and Macroeconomics Aggregates responses to a carbon price shock

Notes: The figure plots the impulse responses to a shock of 1% to the steady state carbon price level.

Figure 6 shows that the carbon-price shock reduces aggregate emissions by 2.5% in the baseline model. Under abatement rigidity, the emission response is hump shaped and have a peak response after 2 quarters at -0.5% which is much closer to the empirical

estimate of Känzig [2023]. This level of abatement rigidity has almost no impact on the macroeconomic responses to the carbon-policy shock compared with the baseline model, as shown in Figure 10. As the emissions response determines the variation in government revenue, the two cases will be further studied when investigating the redistribution of carbon revenues.

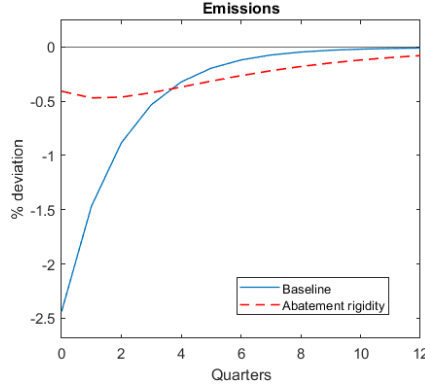


Figure 6: Emissions Response

Notes: The figure plots the impulse responses to a 1% shock to the steady state carbon price level. The red dashed curve represents the response of emissions when the abatement rigidity parameter λ is 0.2.

4.2 Distributional impacts of a carbon price shock

Figure 7 illustrates the impact of a 1% carbon price shock on household consumption, income, and consumption-bundle prices. The cost of a carbon price increase is not borne equally across household groups: the decline in consumption is more than twice as large for hand-to-mouth households (-0.11%) compared with savers (-0.04%), leading to an aggregate response of roughly -0.05%.

The negative effect on household consumption is driven by a combination of a direct effect from higher prices and an indirect effect from lower incomes. Regarding the direct effect, the price index of the consumption bundle increases by 0.052% for hand-to-mouth households and 0.032% for savers, which reflects the discrepancies in energy-expenditure shares. The income response, however, is more pronounced for savers because they reduce the number of hours worked in response to lower wages. They have access to dividends and savings, which allow them to smooth consumption, even if they receive lower labour income. Even if the income response is smaller for hand-to-mouth households, this does not mean they are less affected than savers by the income channel. In contrast to savers, they cannot rely on other income sources and must absorb the wage decrease entirely by reducing consumption.

Table 3 reports the shares of the direct and indirect effects of the carbon price increase on household consumption⁶. The direct price effect accounts for the majority of the decline in consumption for both groups, but indirect effects are much more important for hand-to-mouth households: 49.9% of their consumption decline is explained by lower income, compared with only 15.7% for savers. This result highlights that decreasing wages

⁶I compute these shares using the household budget constraint $p_{i,t}c_{i,t} = r_{i,t}$, where $r_{i,t}$ denotes the resources of household i at time t . Log-linearizing the budget constraint decomposes the variation in consumption into a price effect and a resource effect. Multiplying each term by the steady-state consumption yields impact-level contributions of each effect.

disproportionately affect financially-constrained households, consistent with the findings of Känzig [2023], who shows that indirect effects explain most of the consumption decline of low-income households resulting from a carbon-price shock.

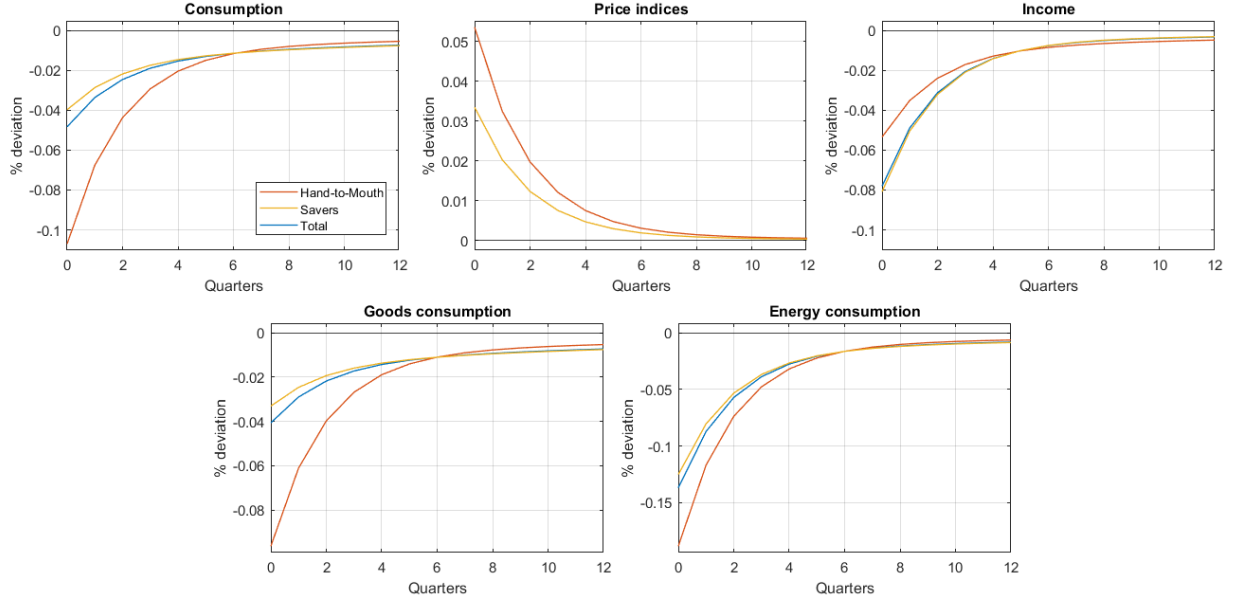


Figure 7: Consumption, prices and income responses to a restrictive carbon policy shock

Notes: The figure plots the impulse responses to a shock of 1% to the steady state carbon price level. The blue line represents aggregate levels. The income variable accounts for the resources available to the agents at each period t , which are only labour income for hand-to-mouth households and labour income, savings and dividends for savers. There is no redistribution of income through transfers in this simulation.

A natural question is which component of consumption between energy or goods is more affected by a rise in the carbon price. I first focus on the estimation of the elasticity of energy demand. The compensated elasticity of substitution is calibrated by the parameter ϵ_x in the model and is equal across household types. I compute the uncompensated elasticity of energy demand with respect to its price, which captures both substitution and income effects⁷. The elasticities are -0.41 for constrained households and -0.27 for savers, which are in line with the estimates of Douenne [2020].

Energy consumption of hand-to-mouth households is more responsive to an energy price change, as their greater energy-expenditure shares and their inability to rely on other income sources amplify the income effect. Figure 7 shows that energy consumption falls by 0.18% for hand-to-mouth households and by 0.12% for savers. Goods consumption also declines about three times as much for hand-to-mouth households as for savers. As the price of consumption goods remains unchanged in the model, this decline comes purely from an income effect, which affects more constrained households.

⁷I use the ratio of the percentage change in energy demand to the percentage change in its price on impact to approximate the log-log elasticity

	Hand-to-Mouth	Savers
Direct effect	50.1 %	84.3 %
Indirect effect	49.9 %	15.7 %

Table 3: Transmission of a carbon price shock on households

Notes: Values are computed at the first period of the shock and are the share of changes in consumption from the direct and indirect effects with respect to the total effect on consumption, on the period of the impact of the shock.

4.3 Horizontal and vertical distributional impacts

Household heterogeneity in energy shares, income structure, and MPC plays an important role for differences in the magnitude of responses to a carbon price increase between groups. To identify which source of heterogeneity matters most, I conduct a counterfactual exercise in which I vary the degree of heterogeneity across households. I compare the baseline model with two alternative versions: one with heterogeneity in energy expenditure shares only, and one with heterogeneity in income incidence only. Removing heterogeneity in income incidence is implemented by perfectly equalizing the resources of hand-to-mouth households to those of savers over the cycle. Figure 8 shows that heterogeneity in income incidence accounts for the largest share of the gap in consumption responses between hand-to-mouth households and savers. When agents differ only in income structure and ability to save, the gap in consumption responses is 0.041%. By contrast, when agents differ only in their energy expenditure shares, the gap is smaller, at 0.02%.

The distributional impact of carbon pricing is thus driven more by differences in income composition and liquidity constraints than by differences in the energy intensity of consumption. But figure 8 also highlights that the horizontal distributional impacts between households from the same income group but with different energy consumption patterns are significant.

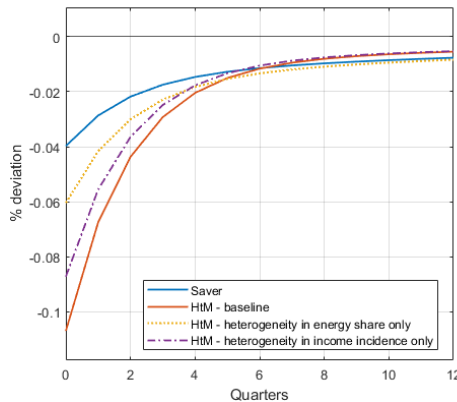


Figure 8: Role of household heterogeneity

4.4 Redistribution of carbon market revenues

How can revenue generated by the carbon market be used to compensate households that bear a higher burden from a carbon-price increase? I address this question by calculating the transfer amount needed to equalize the distributional effects across groups.

Figure 9 compares three cases: the baseline, where revenue from the carbon market is used only to finance government expenditures; a case where 18% of carbon revenue is used as targeted transfers toward hand-to-mouth households; and a case where 18% of carbon revenue is redistributed under abatement rigidity for firms. In this third case, the carbon market generates more revenue as firms must buy more allowances because they cannot drastically reduce their emissions.

An 18% redistribution share of the carbon market revenues allows to equalize the consumption response of hand-to-mouth households with that of savers observed in Figure 7, thereby eliminating disparities in consumption effects due to higher carbon prices. Redistributing income effectively triggers a significant increase in hand-to-mouth consumption because their MPC is very high. When the abatement level is rigid, an 18% redistribution makes the effects of carbon pricing progressive, as hand-to-mouth agents are relatively less affected than savers: they experience a only a 0.02% decrease in consumption, compared with a 0.04% decrease for savers.

This redistribution policy helps reduce aggregate consumption by 0.1% and has little effect on emissions. This is explained by the fact that energy consumption is very inelastic (in terms of the compensated elasticity). Thus, when their income increases, hand-to-mouth households increase their consumption of goods produced by the non-energy sector, which is less carbon-intensive. This is also explained by the fact that hand-to-mouth households make up only a small share of aggregate emissions compared with savers before the redistribution. Thus, the model confirms the intuition that redistributing carbon revenues can mitigate the effects on aggregate consumption and alleviate the distributional impact without compromising emission reductions.

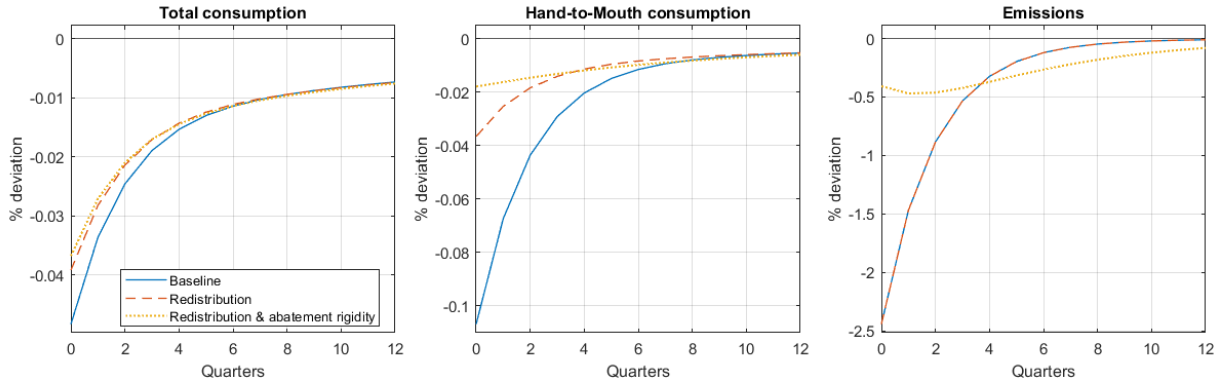


Figure 9: Redistribution of carbon revenues

Notes: The blue line is the baseline response where revenues are only used to finance public expenditures; the red dashed line is the response when 18% of the revenues are redistributed to the hand-to-mouth; the yellow dotted line is the response when abatement is rigid ($\lambda = 0.2$) and 18% of the revenues are redistributed to the hand-to-mouth.

5 Discussion

The results above inform debates on the distributional effects of carbon markets and on how to recycle market-generated revenue. First, they confirm that targeted transfers to households most affected by higher carbon prices can eliminate consumption disparities. More generous, flat lump-sum recycling schemes could even fully offset households' consumption losses and make the policy progressive without weakening market efficiency.

I find that redistributing 18% of market revenue to those most affected—through income losses and higher energy expenditures—is sufficient to restore equality in consumption under carbon pricing. This share is slightly above what is currently envisaged for the European Union's Social Climate Fund (SCF), which is expected to represent between 10% and 15% of total revenues, depending on estimates of future carbon prices, which remain highly uncertain ([Jüngling et al. \[2025\]](#)).

Using revenue for public spending or tax transfers also raises questions about distortions to labour-supply decisions. More generous redistribution can reduce labour supply as households receiving transfers offset income losses. A reduction of labour supply can then further reduce economic activity and output. Alternative uses of carbon revenue—such as subsidies for the implementation of abatement investments for energy suppliers and polluting industries might thus be preferable. By reducing the marginal cost of abatement, such subsidies would encourage firms to decarbonize more, purchase fewer allowances, and limit pass-through to prices and wages. An interesting extension of the model would be to incorporate government-financed abatement subsidies for firms.

An interesting extension would be to implement different wages levels across sectors and study how the distribution of households in these sectors influences the strength of indirect effects on income for each households group. [Känzig \[2023\]](#) finds that households working in sectors sensitive to aggregate demand, such as construction, wholesale, retail trade, or entertainment, face higher income losses. As low-income households are overrepresented in these sectors, the negative effects on their incomes are more significant. Integrating this component into the model could therefore reconcile it with this empirical fact and reveal a greater income response among the low-income households.

6 Conclusion

Through a macroeconomic model calibrated to European data and simulations of an increase in the carbon price, I show that carbon markets have differential effects across household groups. First, I highlight that increasing prices in carbon markets lead to a fall in emissions, but the environmental gains come at the cost of lower economic activity. There is a strong transmission channel from a carbon-price increase, operating through both higher energy prices and lower wages and interest rates, leading to a decrease in consumption and investment. Second, I show that financially constrained low-income households bear a larger share of these economic costs. Their loss of purchasing power through both direct effects on prices and indirect effects on income lowers their consumption more than twice as much as unconstrained households. The indirect income effect impacts low-income households much more and is responsible for the majority of the difference in consumption response compared to higher-income households. I also emphasize significant horizontal distributional effects among households within the same income group but with different energy and fuel-expenditure patterns. Third, I show that redistributing a share of the carbon-market revenues to the most affected groups in

society, such as low-income or rural households with higher fuel-expenditure shares for transport or heating, may be an effective way to reduce the economic costs of carbon pricing. I estimate that the share of revenue from the European carbon market allocated to supporting vulnerable households falls short of what would be needed to equalize the economic impact across household groups. An appropriate redistribution should make it possible to offset the regressivity of the carbon market and strengthen public support for climate policy, without weakening its environmental efficiency.

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A Appendix

A.1 Intratemporal demand schedule

Expressed in real terms, the total expenditures of the household i are $X_{i,t} + P_{e,t}E_{i,t}$. Each period, the household determines its demand schedule for energy and the consumption good by solving the following intratemporal expenditure-minimization problem given a target level of composite consumption $C_{i,t}$:

$$\begin{aligned} \min_{X_{i,t}, E_{i,t}} \quad & X_{i,t} + P_{e,t}E_{i,t} \\ \text{subject to} \quad & C_{i,t} = \left(a_{i,x}^{\frac{1}{\epsilon_x}} X_{i,t}^{\frac{\epsilon_x-1}{\epsilon_x}} + a_{i,e}^{\frac{1}{\epsilon_x}} E_{i,t}^{\frac{\epsilon_x-1}{\epsilon_x}} \right)^{\frac{\epsilon_x}{\epsilon_x-1}} \end{aligned}$$

The lagrangian is:

$$\mathcal{L} = X_{i,t} + P_{e,t}E_{i,t} + \lambda_t[\bar{C}_t - C_{i,t}]$$

The FOC for $E_{i,t}$ and $X_{i,t}$ are:

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial X_{i,t}} &= 1 - \lambda_t C_t^{\frac{1}{\epsilon_x}} a_{i,x}^{\frac{1}{\epsilon_x}} X_{i,t}^{\frac{-1}{\epsilon_x}} = 0 \\ \frac{\partial \mathcal{L}}{\partial E_{i,t}} &= P_{e,t} - \lambda_t C_t^{\frac{1}{\epsilon_x}} a_{i,e}^{\frac{1}{\epsilon_x}} E_{i,t}^{\frac{-1}{\epsilon_x}} = 0 \end{aligned}$$

Dividing the FOC for $E_{i,t}$ by the FOC for $X_{i,t}$ gives:

$$X_{i,t} = \frac{a_{i,x}}{a_{i,e}} \left(\frac{1}{P_{e,t}} \right)^{-\epsilon_x} E_{i,t}$$

Plugging this expression into the consumption aggregator :

$$\begin{aligned} C_{i,t} &= \left(a_{i,x}^{\frac{1}{\epsilon_x}} \left(\frac{a_{i,x}}{a_{i,e}} \left(\frac{1}{P_{e,t}} \right)^{-\epsilon_x} E_{i,t} \right)^{\frac{\epsilon_x-1}{\epsilon_x}} + a_{i,e}^{\frac{1}{\epsilon_x}} E_{i,t}^{\frac{\epsilon_x-1}{\epsilon_x}} \right)^{\frac{\epsilon_x}{\epsilon_x-1}} \\ &= E_{i,t} \left(a_{i,e}^{-\frac{(\epsilon_x-1)}{\epsilon_x}} \frac{a_{i,x}}{P_{e,t}^{1-\epsilon_x}} + a_{i,e}^{\frac{1}{\epsilon_x}} \right)^{\frac{\epsilon_x}{\epsilon_x-1}} \\ &= E_{i,t} a_{i,e}^{-1} \left(\frac{a_{i,x}}{P_{e,t}^{1-\epsilon_x}} + a_{i,e} \right)^{\frac{\epsilon_x}{\epsilon_x-1}} \\ &= E_{i,t} a_{i,e}^{-1} P_{e,t}^{\epsilon_x} P_{e,t}^{-\epsilon_x} \left(\frac{a_{i,x}}{P_{e,t}^{1-\epsilon_x}} + a_{i,e} \right)^{\frac{\epsilon_x}{\epsilon_x-1}} \\ &= E_{i,t} a_{i,e}^{-1} P_{e,t}^{\epsilon_x} (P_{e,t}^{1-\epsilon_x})^{\frac{\epsilon_x}{\epsilon_x-1}} \left(\frac{a_{i,x}}{P_{e,t}^{1-\epsilon_x}} + a_{i,e} \right)^{\frac{\epsilon_x}{\epsilon_x-1}} \\ &= E_{i,t} a_{i,e}^{-1} P_{e,t}^{\epsilon_x} (a_{i,x} + a_{i,e} P_{e,t}^{1-\epsilon_x})^{\frac{\epsilon_x}{\epsilon_x-1}} \\ &= E_{i,t} a_{i,e}^{-1} P_{e,t}^{\epsilon_x} P_{i,t}^{-\epsilon_x} \end{aligned}$$

The demand schedule for energy is then given by:

$$E_{i,t} = a_{i,e} \left(\frac{P_{e,t}}{P_{i,t}} \right)^{-\epsilon_x} C_{i,t}$$

Similarly, we find the demand schedule for consumption goods:

$$X_{i,t} = a_{i,x} \left(\frac{1}{P_{i,t}} \right)^{-\epsilon_x} C_{i,t}$$

A.2 Rigidity of the abatement level

This section introduces the case in which abatement level adjusts gradually and firms are forward-looking about it. In each period, only a fraction $\lambda \in (0, 1]$ of energy producers can reset their abatement level and the remaining $1 - \lambda$ keep last period's level. I present here the case for energy producers, but the same reasoning can be applied to producers of intermediate goods.

Let $\beta \in (0, 1)$ denote the discount factor and define ρ the discounted probability of not being able to reset the abatement:

$$\rho \equiv \beta(1 - \lambda) \in (0, 1).$$

The law of motion for effective abatement follows:

$$\mu_{e,t} = \lambda \mu_{e,t}^{\#} + (1 - \lambda) \mu_{e,t-1},$$

where $\mu_{e,t}^{\#} \in [0, 1]$ is the abatement level chosen by firms that receive a reset opportunity at t . The abatement component of period profits at $t + h$ is

$$\pi_{t+h}(\mu) = -f(\mu)E_{t+h} - \tau_{t+h} \phi_e (1 - \mu) E_{t+h}.$$

Because the abatement level persists until the next reset, firms maximize the expected present value of profits:

$$\max_{\mu \in [0, 1]} \sum_{h=0}^{\infty} \rho^h \mathbb{E}_t[\pi_{t+h}(\mu)].$$

The first-order condition with respect to μ is:

$$\sum_{h \geq 0} \rho^h \mathbb{E}_t[-f'(\mu)E_{t+h} + \phi_e \tau_{t+h} E_{t+h}] = 0.$$

The FOC evaluated at the optimum $\mu = \mu_{e,t}^{\#}$ can be written as follows:

$$f'(\mu_{e,t}^{\#}) = \phi_e \frac{\sum_{h \geq 0} \rho^h \mathbb{E}_t[\tau_{t+h} E_{t+h}]}{\sum_{h \geq 0} \rho^h \mathbb{E}_t[E_{t+h}]}$$

Under a first-order linear approximation around the steady state so that $E_{t+h} = \bar{E} +$

$\tilde{E}_{t+h} \approx \bar{E}$, and given that $\sum_{h \geq 0} \rho^h = \frac{1}{1-\rho}$, we obtain:

$$f'(\mu_{e,t}^\#) = \phi_e (1 - \rho) \sum_{h \geq 0} \rho^h \mathbb{E}_t[\tau_{t+h}] \equiv \phi_e \tilde{\tau}_t.$$

$\tilde{\tau}_t$ here is a weighted forward-looking average of the carbon price:

$$\begin{aligned} \tilde{\tau}_t &= (1 - \rho) \sum_{h \geq 0} \rho^h \mathbb{E}_t[\tau_{t+h}] \\ &= (1 - \rho)(\tau_t + \sum_{h \geq 1} \rho^h \mathbb{E}_t[\tau_{t+h}]) \end{aligned}$$

Given that

$$\begin{aligned} \mathbb{E}_t[\tau_{t+1}] &= (1 - \rho) \sum_{h \geq 0} \rho^h \mathbb{E}_t[\tau_{t+1+h}] \\ &= (1 - \rho) \sum_{h \geq 1} \rho^{h-1} \mathbb{E}_t[\tau_{t+h}] \end{aligned}$$

We have

$$\rho \mathbb{E}_t[\tau_{t+1}] = (1 - \rho) \sum_{h \geq 1} \rho^h \mathbb{E}_t[\tau_{t+h}]$$

We can rewrite $\tilde{\tau}_t$ the forward-looking average of the carbon price, together with the simplified FOC of $\mu_{e,t}^\#$ and the law of motion of effective abatement, to get the new conditions of optimal abatement of firms:

$$\begin{aligned} \tilde{\tau}_t &= (1 - \rho) \tau_t + \rho \mathbb{E}_t[\tau_{t+1}], \\ \tilde{\tau}_t &= \frac{f'(\mu_{e,t}^\#)}{\phi_e}, \quad \mu_{e,t} = \lambda \mu_{e,t}^\# + (1 - \lambda) \mu_{e,t-1}, \end{aligned}$$

A.3 Intermediate goods firm problem

To solve the profit maximization problem of intermediate firms, we first solve the cost minimization problem of choosing production inputs subject to the production function $y_t(j)$. The problem of firm i reads:

$$\begin{aligned} \min_{L_{y,t}(i), K_{y,t}(i), E_{y,t}(i), \mu_{y_i,t}} \quad & W_t L_{y,t}(i) - R_t K_{y,t}(i) - P_{e,t} E_{y,t}(i) - F(\mu_{y_i,t}) - \mathcal{E}_{y_i,t} \tau_t \\ \text{s.t.} \quad & y_t(j) \leq A_t d(s_t) K_{y,t}(i)^\alpha E_{y,t}(i)^\nu L_{y,t}(i)^{1-\alpha-\nu} \end{aligned}$$

The first-order conditions are:

$$\begin{aligned} R_t &= \varrho_{y,t}(i) \alpha \frac{y_t(i)}{K_{y,t}(i)} \\ P_{e,t} &= \varrho_{y,t}(i) \nu \frac{y_t(i)}{E_{y,t}(i)} \\ W_t &= \varrho_{y,t}(i) (1 - \nu - \alpha) \frac{y_t(i)}{L_{y,t}(i)} \\ \tau_t &= \frac{f'(\mu_{y,t})}{\phi_y} \end{aligned}$$

$\varrho_{y,t}(i)$, which is the lagrange multiplier associated to the production function constraint, has the interpretation of the marginal cost of production net of abatement and carbon price costs. As firms are identical, this marginal cost component is identical across firms: $\varrho_{y,t}(i) = \varrho_{y,t}$. Since factor prices are similar across intermediate firms, they purchase capital, labour and energy in similar ratios. The optimality conditions are therefore identical across firms and we can rewrite the FOC above by removing the index i . Aggregate production, and the profit equation of firm i are:

$$\begin{aligned} y_t &= \int_0^1 y_t(i) di \\ \Pi_{y,t}(i) &= \left(\frac{p_t(i)}{P_t} - mc_t(i) \right) y_t(i) \end{aligned}$$

and we can retrieve the expression of the total marginal cost $c_t = mc_t(i) = \varrho_{y,t} + f(\mu_{y,t}) + (1 - \mu_{y,t})\phi_{y,t}\tau_t$.

We can now derive firm's marginal cost and dividend level by solving the firms profit maximization problem:

$$\begin{aligned} \max_{p_t(i)} \quad & \Pi_{y,t}(i) = \left(\frac{p_t(i)}{P_t} - mc_t \right) y_t(i) \\ s.t. \quad & y_t(i) = \left(\frac{p_t(i)}{P_t} \right)^{-\epsilon_y} Y_t \end{aligned}$$

The FOC yields the price level p_t as all firms are identical (i.e. $p_{j,t} = p_t$):

$$\frac{p_t}{P_t} = \frac{\epsilon_y}{\epsilon_y - 1} mc_t$$

The price index $P_t = \left(\int_0^1 p_t(i)^{1-\epsilon_y} di \right)^{\frac{1}{1-\epsilon_y}} = p_t \left(\int_0^1 di \right)^{\frac{1}{1-\epsilon_y}} = p_t$, thus we can rewrite the marginal cost as :

$$mc_t = \frac{\epsilon_y - 1}{\epsilon_y}$$

The dividends of intermediate companies are determined by incorporating the demand

function and using the symmetric equilibrium condition $P_t = p_t$:

$$\begin{aligned} d_t &= \int_0^1 \Pi_{y,t}(i) di \\ &= \int_0^1 \left(\frac{p_t(i)}{P_t} - mc_t \right) \left(\frac{p_t(i)}{P_t} \right)^{-\epsilon_y} Y_t \\ &= (1 - mc_t) Y_t \end{aligned}$$

A.4 Abatement rigidity

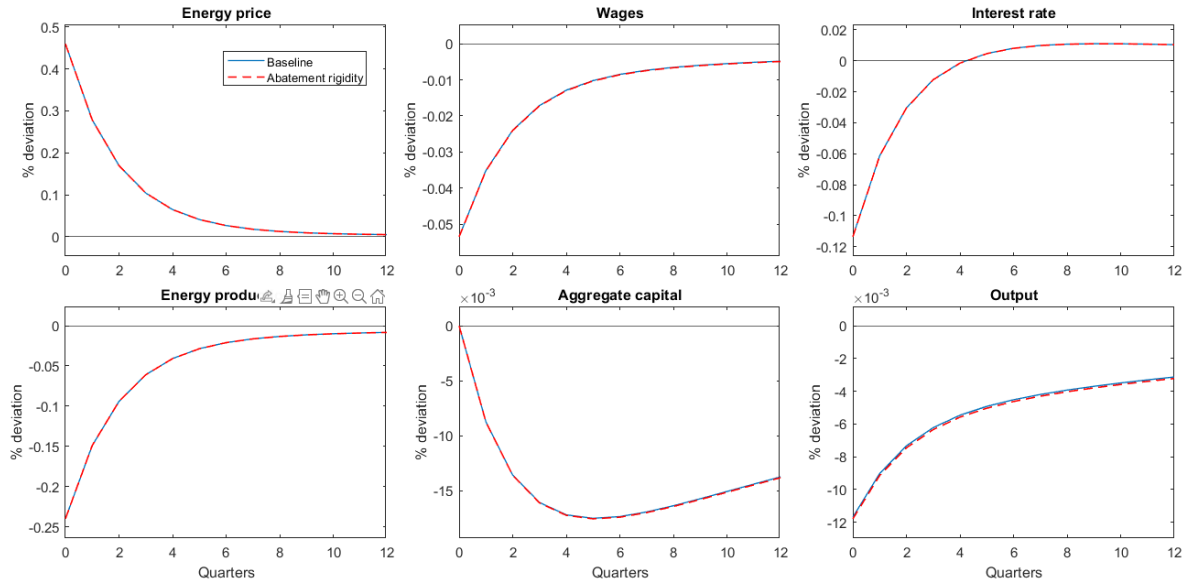


Figure 10: Abatement rigidity and macroeconomic dynamics